ENSIA – 2026-2027
Deep Learning

Lab 4 – Part 2

Experiments of this lab will be performed on Playground:
https://playground.tensorflow.org

Exercise 1:

In this exercise, you will train a simple neural network. Neural nets will give us a way to learn nonlinear models without the use of explicit feature crosses.

You have to use only the two input features X_1 and X_2.
Noise has to be set to 10%.

Task 1: The first model you have to use combines the two input features into a single hidden neuron. The output layer remains with two neurons.

> *Will this model learn any nonlinearities? Run it to confirm your guess?*

Task 2: Try increasing the number of neurons in the hidden layer from 1 to 2, and also try changing from a Linear activation to a nonlinear activation like ReLU.

> *Can you create a model that can learn nonlinearities? Is it capable to model the data effectively?*

Task 3: Try increasing the number of neurons in the hidden layer from 2 to 3, using a nonlinear activation like ReLU.

> *Is it capable to model the data effectively? How does the model quality vary from run to run?*

Task 4: Continue experimenting by adding or removing hidden layers and neurons per layer. Also feel free to change the learning rate, the regularization method and parameters, and other learning settings.

> *What is the smallest number of neurons and layers you can use in order to get a test loss of 0.177 or lower?*
>
> *What is the impact of increasing the model size (the quality of the fit, the speed of the convergence…?*

Exercise 2:

Use the following URL (in which parameters of the data and the model are already set):

https://playground.tensorflow.org/#activation=tanh&batchSize=10&dataset=circle&regDataset=reg-plane&learningRate=0.03&regularizationRate=0&noise=10&networkShape=4,2&seed=0.24342&showTestData=false&discretize=false&percTrainData=50&x=true&y=true&xTimesY=false&xSquared=false&ySquared=false&cosX=false&sinX=false&cosY=false&sinY=false&collectStats=false&problem=classification&initZero=false&hideText=false

The data set contains 10% of noise.

This exercise consists of three related tasks. To efficiently analyze and compare results across the three tasks, run each task in a separate tab.

Task 1: Run the model as given for at least 1000 epochs. Note the following:

- Test loss.
- The difference between the test loss and the training loss.
- The learned weights. (The relative thickness of each line running from FEATURES to OUTPUT represents the learned weight for that feature or feature cross. You can find the exact weight values by hovering over each line.)

Task 2: Increase the regularization rate from 0 to 0.03. Then, run the model for at least 1000 epochs and find answers to the following questions:

- How does the test loss in Task 2 differ from the Test loss in Task 1?
- How does the difference between the test loss and training loss in Task 2 differ from that of Task 1?
- How do learned weights of each feature and feature cross differ from those of Task 1?
- What do your results say about model complexity?

Task 3: Continue modifying the regularization parameters and try to find those that achieve the best performance.

Exercise 3:

In this exercise, you will train a neural net on the spiral data.

You have to use only the two input features X_1 and X_2.
The noise rate has to be set to 0%.

This data set is a noisy spiral. Obviously, a linear model will fail here, but even manually defined feature crosses may be hard to construct.

Task 1: Train the best model you can using just X_1 and X_2. Feel free to add or remove layers and neurons, change learning settings like learning rate, regularization rate, and batch size.

What is the best test loss you can get? How smooth is the model output surface?

Task 2: Even with Neural Nets, feature engineering is sometimes needed to improve the performance. Try adding in additional cross product features or other transformations like $\sin(X_1)$ and $\sin(X_2)$.

Did you get a better model? Is the model output surface smoother?